# Seongyu Kim

seon@mmai.io/kimsk46@kaist.ac.kr seongyukim.github.io
Google Scholar LinkedIn GitHub
Multimodal AI Lab, KAIST

## Research Interest

My research focuses on multimodal AI, with a particular interest in cross-modal alignment spanning diverse sensory modalities — including vision, audio, tactile and olfaction. I explore how self-supervised learning captures correspondences across modalities, aiming to connect and interpret multi-sensory information in real-world environments.

## Education

**Korea Advanced Institute of Science and Technology (KAIST)** Mar 2023 – Present
Integrated M.S./Ph.D. in Electrical Engineering Daejeon, South Korea

- Advisor: Joon Son Chung

**Korea Advanced Institute of Science and Technology (KAIST)** Mar 2018 – Feb 2023
B.S. in Electrical Engineering Daejeon, South Korea

## Publications

**C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis**

[C.3] **See & Sniff: Learning Visuo-Olfactory Representations.**
**Seongyu Kim***, Seungwoo Lee*, Hyeonggon Ryu, Joon Son Chung, Arda Senocak.
European Conference on Computer Vision (**ECCV**) **Oral (Spotlight)**, 2026.

[C.2] **Seeing Through Touch: Tactile-Driven Visual Localization of Material Regions.**
**Seongyu Kim**, Seungwoo Lee, Hyeonggon Ryu, Joon Son Chung, Arda Senocak.
IEEE / CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.

[J.1] **EDNet: A Distortion-Agnostic Speech Enhancement Framework with Gating Mamba Mechanism and Phase Shift-Invariant Training.**
Doyeop Kwak, Youngjoon Jang, **Seongyu Kim**, Joon Son Chung.
IEEE Transactions on Audio, Speech and Language Processing (**TASLP**), 2026.

[C.1] **Seeing Speech and Sound: Distinguishing and Locating Audio Sources in Visual Scenes**
Hyeonggon Ryu*, **Seongyu Kim***, Joon Son Chung, Arda Senocak.
IEEE / CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.

** denotes equal contribution.*

## Honors and Awards

**Outstanding Teaching Assistant Award** 2025
Deep Learning for Visual Understanding

**National Scholarship, KAIST** Mar 2023 – Present
Full tuition

**Dean's List, College of Engineering, KAIST** 2021

## Academic Activities

**Deep Learning for Computer Vision (EE488/EE40034)** Fall 2023, 2024, 2025
Teaching Assistant
- Head TA in Fall 2025

**Speech Recognition Systems (EE738/EE70038)** Spring 2024, 2025
Teaching Assistant
- Head TA in Spring 2025

## Experiences

**Multimodal AI Lab, KAIST** Mar 2022 – Feb 2023
Undergraduate Research Intern South Korea
- Advisor: Joon Son Chung
- Audio-Visual Learning, Sound Source Localization, Speech Recognition

**VIC Lab, KAIST** Dec 2021 – Feb 2022
Undergraduate Research Intern South Korea
- Advisor: Munchurl Kim
- Computer Vision, Super Resolution

**SMILE Lab, KAIST** Jun 2021 – Aug 2021
Undergraduate Research Intern South Korea
- Advisor: Song Min Kim
- Machine Learning, Embedded System

## Skills

**Programming:** Python, PyTorch, C **Languages:** Korean (Native), English (Fluent)

## References

**Joon Son Chung**
Associate Professor, Electrical Engineering
Korea Advanced Institute of Science and Technology (KAIST)
joonsonchung@gmail.com
Relationship: PhD advisor at KAIST

**Arda Senocak**
Assistant Professor, Graduate School of Artificial Intelligence
Ulsan National Institute of Science and Technology (UNIST)
ardasnck@unist.ac.kr
Relationship: Mentor